

Lepromatous Leprosy

In lepromatous leprosy (LL), the absence of cell-mediated immunity (CMI) against *Mycobacterium leprae* allows uncontrolled bacterial replication, leading to widespread, multiorgan involvement. Diffuse dermal infiltration is invariably present, often subclinical, but in diffuse non-nodular LL, it becomes clinically apparent with characteristic features such as:

- Enlargement of the ear lobes
- Widening of the nasal root
- Fusiform swelling of the fingers, which may mimic a rheumatic disorder

Poorly defined nodules are the most common skin lesions, typically up to 2 cm in diameter and symmetrically distributed (Fig. 186-5; see eFig. 186-5.1 in online edition). The skin may fold extensively, resulting in leonine facies, frequently in association with nodular lesions.

Dermatofibroma-like (histoid) lesions are usually multiple, sharply demarcated, erythematous papules or nodules, sometimes coalescing into plaques. Initially described in relapsing cases as "histoid leprosy," these lesions are not uncommon as initial presentations in patients with subpolar lepromatous leprosy (Fig. 186-6; see eFig. 186-6.1 in online edition). Their uniform sharp margins and erythema contrast with the more diffuse lesions seen in typical LL.

Less common skin manifestations include: - Digitate, faintly indurated erythematous patches (Fig. 186-7)

- In light-skinned individuals: may evolve into mild hyperpigmentation
- In dark-skinned individuals: may present as multiple hypopigmented macules due to melanin masking underlying erythema
- Dermal infiltration resembling a shagreen patch (see eFig. 186-7.1 in online edition)

Hair Loss and Neurologic Involvement

- **Hair loss** commonly affects the eyebrows, progressing from lateral to medial or appearing patchy; eyelashes and extremity hair may also be involved (see eFig. 186-7.2 in online edition)
- Scalp involvement is rare, likely due to higher local skin temperature
- **Small-fiber peripheral sensory impairment (S-GPSI)** is frequent and can be severe, leading to debilitating trophic changes in the hands and feet

Histopathology of LL (LLs and LLp)

Both forms share several histologic features:

1. **Nodular lesions** show abundant dermal macrophages (foamy or undifferentiated), sparse lymphocytes, and a thin epidermis overlying a grenz zone
2. **Clinically normal skin** is typically infiltrated; when minimal, infiltration is perivascular
3. **Dense "pseudofollicular" lymphocytic aggregates**, often composed of B cells[22]
4. **Acid-fast bacilli** are readily detected in endothelial cells and Schwann cells
5. **Plasma cells and mast cells** may be increased, though inconsistently
6. **Foreign-body giant cells** may appear in older lesions

Differences between LL subtypes: - In **LLs (subpolar)**: perineurium is laminated but not infiltrated

- In **LLp (polar)**: perineurium remains undisturbed

Indeterminate Leprosy

Indeterminate leprosy is a term variably defined. We adopt Khanolkar's definition[23], which describes it as an early lesion occurring before the host

mounts a definitive immune response—either toward clearance or progression to overt granulomatous disease.